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Inventor(s) Thorne; Daniel et al.

CARBON DIOXIDE-BASED CLEANING SYSTEMS

Abstract

Carbon dioxide-based cleaning systems include a holding tank containing a supply of liquid carbon dioxide (CO.sub.2), a rotary cleaner having an internal drum arranged therein, the rotary cleaner fluidly coupled to the holding tank to receive liquid CO.sub.2 therefrom, the rotary cleaner configured to output CO.sub.2 with contaminants after a cleaning cycle, a compressor arranged to receive gaseous CO.sub.2 and increase a pressure of the gaseous CO.sub.2, and a condenser arranged to receive the increased pressure gaseous CO.sub.2 from the compressor and convert the gaseous CO.sub.2 into liquid CO.sub.2 and direct the liquid CO.sub.2 into the holding tank.

Inventors: **Thorne; Daniel (Terryville, CT), Schuster; Ashley Rose Himmelmann (Chicago, IL), Kaplan; Daniel J. (Melbourne Beach, FL)**

Applicant: **Hamilton Sundstrand Space Systems International, Inc.** (Windsor Locks, CT)

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Background/Summary

BACKGROUND

[0001] The subject matter disclosed herein generally relates to carbon dioxide-based cleaning systems and, more particularly, to carbon dioxide cleaning systems for use in low-, micro-, and zero-gravity environments.

[0002] In current space exploration and travel, supplying clean clothing and cleaning of dirty clothing provides logistical and financial challenges. Currently, there is no way to clean dirty laundry in space. As such, clothing is a consumable that must be brought up to a space station or vehicle, or must be carried onboard. Once the clothing has been used and worn out, the clothing is discarded, and new clothing must be provided. For example, clothing on the International Space Station is supplied to the Station about once a year, with about 68 kg of clothing being launched for each astronaut on the Station. With very high launch costs based on weight (e.g., $10,000/kg), and multiple astronauts, the cost of providing clean clothing can be very expensive, and requires space both on the launch vehicle each time plus storage space onboard the Station for new and old clothing.

[0003] Additionally, as human space exploration expands, such as by way of deep space missions, extended missions on the Moon, to asteroids, Mars, and the like, resupply and replacement clothing cannot be easily or feasibly provided. On such missions, all clothing to be used for the duration of the mission must be launched and then carried by the crew and craft/station. Conventional cleaning systems require water and are designed to operate on Earth, having gravity at play along with access to resources, such as water. In view of this, it may be beneficial to develop systems for cleaning clothing in low-, micro-, and zero-gravity environments without water.

SUMMARY

[0004] According to some embodiments, carbon dioxide-based cleaning systems are provided. The cleaning systems include a holding tank containing a supply of liquid carbon dioxide (CO.sub.2), a rotary cleaner having an internal drum arranged therein, the rotary cleaner fluidly coupled to the holding tank to receive liquid CO**2** therefrom, the rotary cleaner configured to output CO.sub.2 with contaminants after a cleaning cycle, a compressor arranged to receive gaseous CO.sub.2 and increase a pressure of the gaseous CO.sub.2, and a condenser arranged to receive the increased pressure gaseous CO.sub.2 from the compressor and convert the gaseous CO.sub.2 into liquid CO.sub.2 and direct the liquid CO.sub.2 into the holding tank.

[0005] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include that the output CO.sub.2 with contaminants is directed from the rotary cleaner to the compressor after the cleaning cycle.

[0006] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a distillation tank arranged between the rotary cleaner and the compressor, the distillation tank configured to receive at least one of gaseous and liquid CO.sub.2 from the rotary cleaner and separate contaminants from the CO.sub.2 and supply clean gaseous CO**2** to the compressor.

[0007] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include that the distillation tank is coaxially arranged within the rotary cleaner.

[0008] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include that the distillation tank comprises a valve configured to selective fluidly couple a distillation cavity defined within the distillation tank with a cleaning cavity defined within the rotary cleaner and external to the distillation tank.

[0009] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a heating element arranged to provide heat to the distillation tank.

[0010] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a slug tank arranged to receive CO.sub.2 and contaminants captured by the CO.sub.2 from the distillation tank.

[0011] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include that the internal drum is porous to liquid CO.sub.2 and contaminants carried by the liquid CO.sub.2 but prevents clothing from passing out of the internal drum.

[0012] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include that the rotary cleaner comprises a cleaning cavity defined within the internal drum and an annular cavity is defined radially between an external surface of the internal drum and an internal surface of an outer housing of the rotary cleaner.

[0013] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a valve arranged on the rotary cleaner and configured to selectively open to evacuate gas from a cleaning cavity of the rotary cleaner.

[0014] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a pump associated with the valve, the pump configured to provide a motive force to draw the gas from the cleaning cavity of the rotary cleaner.

[0015] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a filter and pump connected to the rotary cleaner, wherein the pump is configured to pull liquid CO.sub.2 from the rotary cleaner and pass it through the filter and then redirect the liquid CO.sub.2 back into the rotary cleaner.

[0016] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a dispenser arranged along a flow path through the pump and the filter, wherein the dispenser is configured to apply at least one of a deodorizer and a detergent to the liquid CO.sub.2 prior to being redirected back into the rotary cleaner.

[0017] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include an auxiliary supply of CO.sub.2 arranged upstream from the compressor, the auxiliary supply of CO.sub.2 configured to provide additional CO.sub.2 into the system to replenish losses from the holding tank.

[0018] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a slug tank arranged to receive CO.sub.2 and contaminants captured by the CO.sub.2 from the rotary cleaner.

[0019] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include a valve configured to fluidly couple a cleaning cavity of the rotary cleaner with an ambient environment.

[0020] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include that the ambient environment is defined within a spacecraft.

[0021] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include that the ambient environment is defined within a space station.

[0022] In addition to one or more of the features described herein, or as an alternative, further embodiments of the carbon dioxide-based cleaning systems may include that the ambient environment is defined within a station on a non-Earth celestial object.

[0023] In addition to one or more of the features described herein, or as an alternative, further

embodiments of the carbon dioxide-based cleaning systems may include that the holding tank is a bellows tank arranged to maintain the liquid CO.sub.2 under pressure and in a liquid state.
[0024] The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, that the following description and drawings are intended to be illustrative and explanatory in nature and non-limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0025] The subject matter is particularly pointed out and distinctly claimed at the conclusion of the specification. The foregoing and other features, and advantages of the present disclosure are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:
[0026] FIG. **1** is a schematic diagram of a closed-loop CO.sub.2 cleaning system for use in low-, micro-, and zero-gravity environments, in accordance with an embodiment of the present disclosure;
[0027] FIG. **2** is a schematic diagram of another embodiment of a closed-loop CO.sub.2 cleaning system in accordance with an embodiment of the present disclosure;
[0028] FIG. **3**A is a front elevation schematic of a rotary cleaner in accordance with an embodiment of the present disclosure;
[0029] FIG. **3**B is a side elevation view of the rotary cleaner of FIG. **3**A;
[0030] FIG. **4** is a schematic diagram of another embodiment of a closed-loop CO.sub.2 cleaning system in accordance with an embodiment of the present disclosure;
[0031] FIG. **5** is a schematic diagram of an embodiment of an open-loop CO.sub.2 cleaning system in accordance with an embodiment of the present disclosure; and
[0032] FIG. **6** is a schematic diagram of another embodiment of an open-loop CO.sub.2 cleaning system in accordance with an embodiment of the present disclosure.
DETAILED DESCRIPTION
[0033] As shown and described herein, various features of the disclosure will be presented. Various embodiments may have the same or similar features and thus the same or similar features may be labeled with similar reference numerals and/or description thereof may be omitted in certain later described embodiments for conciseness. Although similar reference numbers may be used in a generic sense, various embodiments will be described and various features may include changes, alterations, modifications, etc. as will be appreciated by those of skill in the art, whether explicitly described or otherwise would be appreciated by those of skill in the art. Further, it will be appreciated that, unless otherwise stated, features from the various separately described embodiments may be combined in various combinations and each embodiment is not intended to be mutually exclusive from features of other embodiments described herein and/or mutually exclusive from other features and components not explicitly described.
[0034] As human space exploration continues and the time and distance of the duration in space continues, clean clothing supply will become an issue that must be addressed. Currently, there is no way to clean dirty laundry in space, and as a result, worn out and used clothing is discarded with trash and other waste. Such a solution has been feasible for low Earth orbit stations (e.g., International Space Station), as the Station is relatively close to Earth, resupply missions happen regularly, and the Station is generally government supported, resulting is relatively high budget that can absorb the cost of regular clothing replacement and resupply. For example, 150 pounds or 68 kg of fresh clothes must be replenished on crewed missions to the International Space Station.

Assuming a cost of about \$10,000/kg, this results in a cost of about \$680,000 of launch cost per crew member per year.

[0035] Future deep space crewed missions will not be able to throw anything away, and so a solution for cleaning dirty clothes is needed. Moreover, as more commercial space exploration expands, reducing the costs of launch and resupply (e.g., by reducing the amount of clothing launched) can provide significant benefits. Conventional clothing cleaning relies upon the use of water. However, water is a scarce and very limited resource when in space, whether on a spacecraft, a station, or a base/station that is on a celestial body (e.g., the Moon, Mars, asteroid, etc.). Additionally, because such space-based cleaning operations must necessarily take place in non-Earth-based gravity environments (e.g., low-, micro-, or zero-gravity environments), cleaning systems must be configured to operate in such space environments, without the assistance of gravity. As used herein, the term "microgravity" will be used to encompass low-gravity, micro-gravity, and zero-gravity environments. Microgravity environments, as intended herein, includes environments including, but not limited to, orbits about a celestial body, open space travel or transit between celestial bodies, and/or surface environments on celestial bodies having gravity lower than that of Earth (e.g., the Moon, Mars, asteroids, moons, planetesimal, etc.).

[0036] In view of the above and other considerations, embodiments of the present disclosure are directed to space-based clothing cleaning systems. In accordance with some embodiments of the present disclosure, carbon dioxide is used as cleaning fluid and replaces the need for water to be used for cleaning of clothing. This results in a waterless cleaning solution for both low Earth orbit and deep space missions where water may be scarce. At a minimum, such systems may enable reduced launch and resupply costs by enabling cleaning of clothing and thus reuse thereof, avoiding the need for resupply. Furthermore, such systems may reduce the total waste generation of a space mission. Further advantages and benefits of embodiments of the present disclosure will be apparent in view of the following discussion and description of various non-limiting embodiments of the present disclosure. Although a number of different embodiments will be described herein, it will be appreciated that features of the various embodiments may be interchanged and/or exchanged to result in one or more embodiments not expressly described but rather are encompassed by the spirit and scope of the disclosure. Further, additional features and structures may be incorporated into embodiments without departing from the scope of the present disclosure.

[0037] In accordance with embodiments of the present disclosure, carbon dioxide (CO.sub.2) based cleaning systems for cleaning clothes are provided. The cleaning systems use liquid CO.sub.2. Contaminants from dirty clothes are removed by the low viscosity and surface tension of the liquid CO.sub.2. Once the cleaning process is completed, the pressure is reduced and the liquid CO.sub.2 will be converted to a gas state, which can allow for scrubbing and removal of dirt, debris, and waste that is removed from the clothing during the cleaning process. The gas CO.sub.2 may be reclaimed and the waste separated therefrom. The gas CO.sub.2 may then be reused for subsequent cleaning processes or cycles. Embodiments of the present disclosure provide for a waterless solution for cleaning clothes in space. Further, in accordance with some embodiments, the cleaning process may occur at ambient (e.g., room) temperatures within an occupied space (e.g., cabin, crew quarters, etc. that are part of a vehicle, station, etc.). By using liquid CO.sub.2 and operating at room temperature, clothing may be dried rapidly after a cleaning cycle or process without the application of heat or other external inputs/operations. The use of liquid CO.sub.2 can result in cleaning of clothes that is more gentle than conventional water-based washing, which can result in increased life/longevity of clothing. This can result in extending the usable life of clothing, which in turn can reduce the total amount of clothing that is launched and carried aboard spacecraft and/or space-based stations and the like.

[0038] As noted, embodiments of the present disclosure are directed to waterless CO.sub.2-based cleaning systems. Such systems may be closed-loop systems, where the CO.sub.2 is recaptured, reclaimed, or otherwise recycled within the system. In some such closed-loop systems, a supplemental supply of CO_2 may be provided to ensure that necessary levels of CO_2 are maintained within the system and to provide a replenishment of CO_2 if losses occur, such as through evaporation, capture and disposal with waste removed from clothing, or the like. In other embodiments, the systems may be open-loop, where the CO_2 and waste are disposed of directly, without reclaiming of the CO_2. Further, some combination of closed- and open-loop may be employed, where a system typically operates as a closed-loop system, but may be converted to open-loop to dispose of used CO_2 and then a resupply of clean CO_2 may be provided, and then converted back to a closed-loop system. In some embodiments, additional chemicals and/or cleaning agents may be used to provide additional cleaning and/or deodorizing to clothes and/or of the system itself. In some embodiments, optional distillers and/or slug tanks may be employed to improve removal of residues, dirt, debris, and waste. Furthermore, it will be appreciated that, beyond what is shown and described herein, the systems may include additional valves, sensors, access ports, quick disconnects, fittings, pressure relief hardware, pressurization hardware, and/or purge hardware may be added to systems described herein. Such additional components may be used to aid in maintenance operations, cleaning operations, or the like, of such systems, as will be appreciated by those of skill in the art and in view of the teachings herein.

[0039] Referring to FIG. **1**, illustrated is a schematic diagram of a cleaning system **100** in accordance with an embodiment of the present disclosure. The cleaning system **100** may be used onboard a spacecraft, space station, non-Earth station or base, or the like, which is subject to a microgravity environment. The cleaning system **100** of FIG. **1** is a closed-loop configuration that uses carbon dioxide (CO_2) for cleaning of clothing. The cleaning system **100** includes a holding tank **102** which is configured to contain liquid CO_2. The holding tank **102** may be a pressure tank, a bellows tank, a variable volume container, or the like, as will be appreciated by those of skill in the art. The holding tank **102** may be configured to ensure a high pressure is maintained such that CO_2 that is within the holding tank **102** is maintained in liquid form. In a flow direction along a flow path **104** from the holding tank **102** is a first pump **106** and a rotary cleaner **108** for receiving both laundry through a door **110** and liquid CO_2 from the primary flow path **104**. The rotary cleaner **108** includes an internal drum **109** that is arranged within the rotary cleaner **108** and provides a mechanism to allow for separation of the liquid CO_2 from clothes during a cleaning operation of the cleaning system **100**. Output from the rotary cleaner **108** may be liquid and/or gaseous CO**2**. Downstream from the rotary cleaner **108**, along the flow path **104**, may be an optional first filter **112** and optional second pump **114**, and then a distillation tank **116**. Continuing along the flow path **104** may be an optional second filter **118**, a compressor **120**, and a condenser **122**, before returning to the holding tank **102**.

[0040] The cleaning system **100** may include additional components, which may be optional in this configuration. For example, as shown, a fluid connection from the rotary cleaner **108** may be provided to fluidly connect to an occupied space **124** (e.g., a cabin or crew quarters) and/or fluidly connect to an air outlet **126** which may include a third pump **128**. Additionally, an auxiliary supply **130** may be provided to add additional CO_2 into the cleaning system **100**. Furthermore, the condenser **112** may be arranged along the flow path **104**, with the flow path **104** providing the CO_2 as a first working fluid within the condenser **122** and a coolant loop **132** providing a second working fluid within the condenser **122**, and thus remove heat from the CO_2 as it passes through the condenser **122**, as described herein. As shown and in this non-limiting configuration, the flow of CO_2, whether liquid or gas, through the cleaning system **100** may be controlled by a set of valves V_1-V_9. The valves V_1-V_9 may be passive (e.g., one-way valves, check valves, or the like) or may be active valves (e.g., solenoid valves, electronic valves, or the like), and each of the valves V_1-V_9 may be configured for manual or automated operation, depending on the specific configuration of the cleaning system **100**. In some embodiments, a controller **134** may be provided for control and operation of the cleaning system **100**, with the controller **134** operably connected to some or all of the valves V_1-V_9.

[0041] An example of operation of the cleaning system **100**, in accordance with a non-limiting embodiment, will now be described. A user will open the door **110** to the rotary cleaner **108** and place soiled clothes into the internal drum **109**. The internal drum **109** may be a perforated container or structure that is configured to rotate with rotation of the rotary cleaner **108**. That is, the internal drum **109** and the rotary cleaner **108** may be commonly driven or may be a single structure that moves (rotates) as a single body. The structure of the internal drum **109** will contain the clothes as the internal drum **109** and rotary cleaner **108** are rotated. The internal drum **109** is perforated such that liquid CO.sub.2 may pass through apertures, holes, perforations, or the like, and flow to a space external to the internal drum **109** and be collected on an internal surface of the rotary cleaner **108**. With the clothes placed in the internal drum **109**, the user will then close the door **110**. When the door **110** is closed it will form a sealing engagement with the rotary cleaner **108** (or a housing thereof) or a sealing operation may be performed, such as a locking mechanism that provides a fluid seal at the door **110** to define a sealed internal volume within the rotary cleaner **108**. In accordance with embodiments of the present disclosure, the rotary cleaner **108** is configured as a rotating centrifuge or rotary phase separator. This configuration enables cleaning of clothes in the absence of gravity and causes the separation of liquids and gasses within the rotary cleaner **108**. Accordingly, embodiments described herein are directed to systems that are uniquely suited for a microgravity environment.
[0042] With the door **110** closed and the rotary cleaner **108** sealed, the valves V.sub.1-V.sub.9 may all be closed (if not already closed). When the clothes are inserted into the internal drum **109**, and the door **110** is open, the internal volume of the rotary cleaner **108** will be filled with air and reach ambient (occupied space) pressure. For operation of the rotary cleaner **108** to perform a cleaning cycle with liquid CO.sub.2, the air within the rotary cleaner must be removed. As such, the ninth valve V.sub.9 will be opened. With the ninth valve V.sub.9 open, the third pump **128** may be operated to draw down the pressure within the rotary cleaner **109** and extract the air therefrom. For example, the third pump **128** may be a vacuum pump that is used to remove the air from the rotary cleaner **108** and direct the extracted air to the air outlet **126**. The air outlet **126** may provide fluid communication to an occupied space (e.g., cabin, crew quarters, etc.) and/or an air processing and filtration system. In other configurations, the third pump **128** may be omitted, and the ninth valve V.sub.9 may be used to expose the internal volume of the rotary cleaner **108** to the vacuum of space, thus pulling the air within the internal volume of the rotary cleaner **108** out. Once the air has been removed from the rotary cleaner **108**, the ninth valve V.sub.9 will close.
[0043] Next, a fill operation is performed. In this operation, the first valve V.sub.1 is opened and the first pump **106** is activated. The first pump **106** will operate to direct a flow of liquid CO.sub.2 from the holding tank **102** through the first valve V.sub.1 and into the internal volume of the rotary cleaner **108**. As the liquid CO.sub.2 enters the internal volume of the emptied rotary cleaner **108**, the CO.sub.2 will boil into gas. As the CO.sub.2 is continued to be pumped into the rotary cleaner **108** by the first pump **106**, the pressure will increase. When the gaseous form of CO.sub.2 within the rotary cleaner **108** hits a threshold pressure level (e.g., about 700 psi), additional CO.sub.2 that is input into the rotary cleaner **108** will remain in liquid form. Once at or above the threshold pressure level, the first pump **106** will continue to pump liquid CO.sub.2 from the holding tank **102** into the rotary cleaner **108** until a specific, predetermined, or threshold fill level is reached. The fill level may be set based on various criteria, which may be load-specific, such that excess waste of materials and energy can be minimized. For example, sensors located within the rotary cleaner **108** may be used to measure the fill levels with the rotary cleaner **108**. In other configurations, a preset or predetermined fill level or amount of CO.sub.2 may be introduced into the rotary cleaner **108**, which may be controlled by flow rate sensors, valves, pumps, and the like.
[0044] The introduction of CO.sub.2 into the rotary cleaner **108** may be achieved by one or more input nozzles, which may be arranged about a periphery of the rotary cleaner **108**, at an end (e.g., opposite the door **110**), and/or through a dispensing shaft arranged axially within the rotary cleaner

108 and the internal drum **109**. When a predetermined or threshold amount of liquid CO.sub.2 has filled the rotary cleaner **108**, the first pump **106** is turned off and the first valve Vi is closed. At this point, the rotary cleaner **108** is fluidly sealed, with clothes held within the internal drum **109** and the rotary cleaner **108** filled with liquid CO**2**.

[0045] Next, a cleaning cycle is performed. During the cleaning cycle, the liquid CO.sub.2 and the dirty clothes will mix and interact. The low viscosity and surface tension of the liquid CO.sub.2 provide a solvent which binds to dirt, debris, contaminants, soiling, and the like that is on the clothes. The rotary cleaner **108** is rotated by a driving mechanism, such as a rotary shaft or the like. As noted above, the internal drum **109** and the rotary cleaner **108** rotate in tandem, and as the internal drum **109** and rotary cleaner **108** rotate, the liquid CO.sub.2 will be forced radially outward through the clothes, through the holes in the internal drum **109**, and collect on an internal surface of the rotary cleaner **108** that is external to the internal drum **109**. In some configurations, an agitator or other element, structure, or feature may be arranged within the internal drum **109** to cause additional interaction between the clothing and the liquid CO.sub.2. In accordance with some embodiments, the cleaning cycle may be performed for a predetermined amount of time (e.g., cleaning cycle time).

[0046] When the cleaning cycle is complete (or at the same time), the distillation tank **116** will be pressurized. In this operation, the second valve V.sub.2 is opened and the first pump **106** is operated. Because the first, third, fourth, eighth, and ninth valves V.sub.1, V.sub.3, V.sub.4, V.sub.8, V.sub.9 are closed, the rotary cleaner **108** remains fluidly isolated. Accordingly, as the first pump **106** is operated, liquid CO.sub.2 from the holding tank **102** may be directed through the second valve V.sub.2 and into the distillation tank **116**. Similar to pressurizing the rotary cleaner **108**, the first pump **106** is operated for sufficient time to bring the pressure up within the distillation tank **116** such that liquid CO.sub.2 is formed and then maintained therein (e.g., about 700 psi). As such, when additional CO.sub.2 is added to the distillation tank **116**, after bringing the pressure up, it will either convert to or remain as liquid CO.sub.2. With the distillation tank **116** at or above the threshold pressure, the second valve V.sub.2 will close.

[0047] As noted, the cleaning cycle may be performed before or during the pressurization of the distillation tank **116**. When the distillation tank **116** is brought to pressure and the cleaning cycle is complete, the rotary cleaner **108** may continue to be rotationally driven. That is, after the cleaning cycle is complete, the rotary cleaner **108** will continue to spin.

[0048] With the rotary cleaner **108** spinning, the dirty liquid CO.sub.2 within the rotary cleaner **108** may be drained. To drain the rotary cleaner **108**, the distillation tank **116** is rotated, such as by a drive shaft or the like, as will be appreciated by those of skill in the art. With the distillation tank **116** spinning, the third valve V.sub.3 is opened and the second pump **114** is activated. The second pump **114** is arranged between the rotary cleaner **108** and the distillation tank **116** and is arranged to cause the dirty liquid CO.sub.2 within the rotary cleaner **108** to flow into the distillation tank **116**. When the liquid CO.sub.2 has been removed from the rotary cleaner **108** and directed into the distillation tank **116**, the third valve V.sub.3 will close. As shown, the flow path **104** between the rotary cleaner **108** and the distillation tank **116** may include the first filter **112**. The first filter **112** may be an optional filter to remove debris such as lint or the like, and/or may be configured to pre-treat the CO.sub.2 prior to being supplied into the distillation tank **116**. As the distillation tank **116** is rotated, any contaminates that were picked up by the liquid CO.sub.2 during the cleaning cycle will be removed from the CO.sub.2. For example, while the distillation tank **116** is rotated, the pressure may be reduced and the temperature may be optionally increased, resulting in the liquid CO.sub.2 to boil so that the CO.sub.2 gas/vapor and contaminants are separated. The contaminants may then collect in the distillation tank **116** and the CO.sub.2 gas/vapor may then be recondensed as clean liquid CO.sub.2.

[0049] That is, with the CO.sub.2 held within the distillation tank **116**, a CO.sub.2 recapture operation may be performed. In the CO.sub.2 recapture operation, the fifth valve V.sub.5 and the

sixth valve V.sub.6 are opened and the compressor **120** is activated. The compressor **120** will generate a low pressure to draw CO.sub.2 out of the distillation tank **116** and direct the CO.sub.2 to the condenser **122**. The CO.sub.2 extracted from the distillation tank **116** may be in gaseous form, and thus it must be converted back to liquid form prior to completing the closed-loop operation and suppling the CO.sub.2 back into the holding tank **102**. Accordingly, as the compressor **120** and the condenser **122** operate, the pressure of the gaseous CO.sub.2 will be increased in the compressor **120** and the temperature decreased in the condenser **122**, cause the gaseous CO.sub.2 to condense and phase change to liquid CO.sub.2 which is supplied back into the holding tank **102**. The temperature of the CO.sub.2 may be reduced within the condenser **122** by means of the second working fluid within the coolant loop **132** acting as a heat sink. The coolant loop **132** may be part of other systems of the spacecraft/station or may be a dedicated coolant loop for the cleaning system **100**.
[0050] In this operation, the compressor **120** is operated until the distillation tank **116** is brought to the same pressure as the rotary cleaner **108**. When the distillation tank **116** and the rotary cleaner **108** at the same pressure, the fourth valve V.sub.4 is opened. The compressor **120** will continue to operate and pull CO.sub.2 gas out of the rotary cleaner **108** and the distillation tank **116** until all gaseous CO.sub.2 has been removed. Accordingly, by running the compressor **120** in this operation, CO.sub.2 that is upstream from the compressor **120** (up to the rotary cleaner **108**) will be pulled through the flow path **104** and supplied back to the holding tank **102**. In some configurations, the compressor **120** may be operated to pull additional CO.sub.2 into the cleaning system **100** from the auxiliary supply **130**. To perform the replenishment operation, the seventh valve V.sub.7 may be opened, and the pressure differential generated by the compressor **120** may pull liquid or gaseous supplemental CO.sub.2 from the auxiliary supply **130** and the supplemental CO.sub.2 may be provided through the compressor **120** and condenser **122** to add additional liquid CO.sub.2 to the holding tank **102**.
[0051] With the CO.sub.2 removed from the rotary cleaner **108** and the distillation tank **116**, the compressor **120** will be turned off, and the fifth valve V.sub.5 will be closed. In some embodiments, the ninth valve V.sub.9 may then be opened to expose the rotary cleaner **108** to vacuum or the third pump **128** is operated to remove any trace amounts of CO.sub.2 that were not removed during the prior described operation. If such venting is performed by operation of the valve V.sub.9, after such venting, the valve V.sub.9 will then be closed. Whether the valve V.sub.9 is operated or not, next the valve V.sub.4 is closed and the rotary cleaner **108** and the distillation tank **116** stop rotating.
[0052] At this point, the clothes within the internal drum **109** of the rotary cleaner **108** will be clean, after having the liquid CO.sub.2 bind, collect, and remove the contaminants. Next, the rotary cleaner **108** must be re-pressurized to ambient pressures to match the external environment of the occupied space and allow for a user to access the inside of the rotary cleaner **108**. Accordingly, in this operation, the valve V.sub.8 will be opened and air from the occupied space **124** may enter the rotary cleaner **108**. This may occur passively because the rotary cleaner **108** was vented and potentially exposed to vacuum, resulting in a low pressure volume that will pull air from the occupied space into the rotary cleaner **108**. In some configurations, an optional pump or the like may be used to assist in this re-pressurization step. The user may then open the door **110** to remove clean clothes from the internal drum **109** of the rotary cleaner **108**.
[0053] Referring now to FIG. **2**, illustrated is a schematic diagram of a cleaning system **200** in accordance with an embodiment of the present disclosure. The cleaning system **200** may be used onboard a spacecraft, space station, non-Earth station or base, or the like, which is subject to a microgravity environment. The cleaning system **200**, similar to that of FIG. **1**, is a closed-loop configuration that uses carbon dioxide (CO.sub.2) for cleaning of clothing. The cleaning system **200** includes a holding tank **202** which is configured to contain liquid CO.sub.2. In a flow direction along a flow path **204** from the holding tank **202** is a first pump **206** and a rotary cleaner **208** for

receiving both laundry through a door **210** and liquid CO.sub.2 from the holding tank **202**. The rotary cleaner **208** includes an internal drum that is arranged within the rotary cleaner **208** similar to that shown and described above and further described with respect to FIGS. **3**A-**3**B. In this configuration, a distillation tank **212** is arranged within the rotary cleaner **208**. The distillation tank **212** may be arranged radially inward from the internal drum of the rotary cleaner **208** and may be configured to be rotationally driven in tandem or simultaneously with the rotary cleaner **208**. That is, the distillation tank **212** may be arranged coaxially with the rotary cleaner **208**. The distillation tank **212** may be maintained selectively fluidly separate from the rest of the cleaning cavity of the rotary cleaner **208**. Output from the rotary cleaner **208** may be liquid and/or gaseous CO.sub.2. It will be appreciated that in other configurations, a commonly mounted or coaxially arranged distillation tank may be provided that is not arranged within the rotary cleaner **208**, but rather may be arranged to rotate on the same axis. The resulting system may be substantially similar to that of the cleaning system **100** shown in FIG. **1**.

[0054] With the distillation tank **212** arranged within the rotary cleaner **208**, the flow paths of the cleaning system **200** are different from that of the cleaning system **100** of FIG. **1**. For example, a first output of the rotary cleaner **208** may be directed to a second pump **214**, and then to a first filter **216** and a dispenser **218** along a first output path of the flow path **204** (e.g., during a cleaning cycle). A second output of the rotary cleaner **208** may be directed to the lower pressure distillation tank **212** (e.g., from valves V.sub.3, V.sub.4). After processing in the distillation tank **212**, contaminants may be directed through a flow control element **224**, and into a slug tank **222**, a second filter **220**, and then to the first filter **216**, and the dispenser **218** along a first output path of the flow path **204** (e.g., optionally during a cleaning cycle or when the slug tank **222** is full). It will be appreciated that the slug tank **222** may be an optional component that is used to aid the distillation tank **212** in separation of contaminants from the CO.sub.2. Then, gaseous CO.sub.2 will be directed into a gaseous holding tank **226**. From the gaseous holding tank **226**, which may be supplemented with additional CO.sub.2 from an auxiliary supply **228**, the gaseous CO.sub.2 can be converted back into liquid CO.sub.2 by being compressed within a compressor **230** and cooled within a condenser **232** before being supplied back into the holding tank **202**. Similar to the embodiment of FIG. **1**, the condenser **232** may be fluidly coupled to a coolant loop **234**. The cleaning system **200** further includes a venting assembly for the rotary cleaner **208**. The venting assembly includes a fourth pump **236** that may be fluidly coupled between the rotary cleaner **208** and an air outlet **238** (or the vacuum of space) and/or the rotary cleaner **208** may be fluidly connected to an occupied space **240**, similar to the configuration of FIG. **1**.

[0055] As shown in this non-limiting configuration, the flow of CO.sub.2 along the flow path **204**, whether liquid or gas, through the cleaning system **200** may be controlled by a set of valves V.sub.1-V.sub.12. The valves V.sub.1-V.sub.12 may be passive (e.g., one-way valves, check valves, or the like) or may be active valves (e.g., solenoid valves, electronic valves, or the like), and each of the valves V.sub.1-V.sub.12 may be configured for manual or automated operation, depending on the specific configuration of the cleaning system **200**. In some embodiments, a controller **242** may be provided for control and operation of the cleaning system **200**, with the controller **242** operably connected to some or all of the valves V.sub.1-V.sub.12.

[0056] In operation, a user will open the door **210** of the rotary cleaner **208** and place soiled or dirty clothes within the internal drum of the rotary cleaner **208**. The door **210** is then closed and sealingly engages to provide a fluid seal around the door **210**. All valves V.sub.1-V.sub.12 are closed, and then the valve V.sub.11 or the valve V.sub.12, at an outlet to the rotary cleaner **208** and arranged between the rotary cleaner **208** and the occupied space **240** and the air outlet **238**, respectively, is opened. In some configurations, the fourth pump **236** may be arranged between the rotary cleaner and each of the valves V.sub.11 and V.sub.12. The fourth pump **236** may be operated to remove air from the rotary cleaner **208** and direct air into the occupied space **240** and/or to an air processing and filtration system. In some embodiments, the rotary cleaner **208** may be exposed to a low pressure external environment, such as the vacuum of space, which may be assisted by the fourth pump **236**. The process is used to evacuate any air from the cleaning cavity of the rotary cleaner **208**. Once the air is removed, the valves V.sub.11 and/or V.sub.12, which may have been opened, will be closed.

[0057] Next, a fill operation is performed. It is noted that because this is a cyclical process, during the fill operation, the distillation tank **212** may already be filled from a prior cycle, or pre-filled during an initial installation and set up. The distillation tank **212** will contain liquid CO.sub.2 at about 700 psi. During the fill operation, valves V.sub.10 and V.sub.6 are opened and the rotary cleaner **208** is filled with CO.sub.2 received from the gaseous holding tank **226**. The drawing of the CO.sub.2 from the gaseous holding tank **226** and into the rotary cleaner **208** may be caused by a pressure differential achieved during the initial evacuation of air from rotary cleaner **208**. The pressure within the rotary cleaner **208** may be monitored to ensure that the pressure reaches at least 75 psi. This operation may be performed to ensure that dry ice does not form within the rotary cleaner **208** and/or on the clothes within the rotary cleaner **208** and/or prevent thermal shock to the system **200** and components thereof. This step may be optional, but it may be used to ensure that the CO.sub.2 within the rotary cleaner **208** is maintained in liquid form.

[0058] Next, valves V.sub.1, V.sub.2 are opened and the first pump **206** is activated. As the first pump **206** is operated, CO.sub.2 from the holding tank **202** will be supplied into the cleaning cavity of the rotary cleaner **208**. The liquid CO.sub.2 will enter the low pressure rotary cleaner **208** and boil and transition into a gaseous state. As CO.sub.2 continues to be supplied into the rotary cleaner **208**, the pressure within the rotary cleaner **208** will increase. The pressure will reach a threshold value (e.g., about 700 psi) where additional input CO.sub.2 will not boil, and thus will fill the rotary cleaner **208** as a liquid. That is, once the rotary cleaner **208** reaches the threshold pressure, the first pump **206** will continue to operate to direct liquid CO.sub.2 from the holding tank **202** into the rotary cleaner **208** to reach a specified fill level and/or pressure level. Once the fill level is reached, valve V.sub.1 is closed and valve V.sub.2 remains open. Next valve V.sub.3 is opened and the second pump **214** is operated.

[0059] This configuration results in a closed loop cycle from the rotary cleaner **208**, through valve V.sub.3, through the second pump **214**, through the first filter **216**, through the dispenser **218**, through valve V.sub.2, and then back into the rotary cleaner **208**. The first filter **216** may be a lint or other debris filter to remove particulates or the like from the cycled CO.sub.2. The dispenser **218** may be an optional component that may dispense detergent, deodorizer, or the like into the flow stream of CO.sub.2 to promote better cleaning and/or to deodorize and/or remove odors from the CO.sub.2 and/or clothes as the CO.sub.2 with detergent/deodorizer (in this arrangement) is supplied back into the rotary cleaner **208** to continue cleaning the clothes within the internal drum of the rotary cleaner **208**. The cleaning cycle is thus performed to remove soil, dirt, and the like from the clothes by operation of the liquid CO.sub.2 being forced through the clothing as the rotary cleaner **208** rotates, as described above. Further, agitation of the clothing may be achieved through the rotation of the internal drum and/or other features, such as blades, vanes, ribs, protrusions, or the like arranged on the internal surface of the internal drum and/or by operation of an injection nozzle or the like for spraying the liquid CO.sub.2 at the clothing in a radially outward direction (i.e., from the center of the internal drum and radially outward therefrom).

[0060] During the agitation process, the pressure and temperature in the rotary cleaner **208** will rise so that the warmer temperature of the liquid CO.sub.2 surrounding the distillation tank **212**, arranged within the rotary cleaner **208**, will cause the liquid CO.sub.2 inside the distillation tank **212** (e.g., from a previous cleaning cycle) to boil. An optional heating element may be provided to introduce additional heat to the distillation tank **212** to aid in the processing that occurs within the distillation tank **212**. For example, as the liquid CO.sub.2 within the distillation tank **212** vaporizes during the cleaning cycle and/or application of heat from the optional heater, soil, dye residue, and other contaminants are left behind. Because the distillation tank **212** is rotated, with the rotary cleaner **208**, the liquid and solid contaminants contained in the CO.sub.2 will be collected on the internal surface of the distillation tank **212** and gaseous CO.sub.2 will be substantially maintained radially inward therefrom. The rotary cleaner **208** will continue to spin through the remainder of the cleaning cycle.

[0061] Next, the distillation tank **212** will be depressurized. The depressurization of the distillation tank **212** is initiated by opening valve V.sub.4 and the flow control element **224** is operated. In some embodiments, the flow control element **224** may be a valve, and in other configurations, the flow control element **224** may be a pump or other fluid motive driver or the like (e.g., a third pump). The flow control element **224** will cause any residue contained within the distillation tank **212** to be extracted and conveyed into the slug tank **222**. The slug tank **222** may be a pressure vessel and, in some embodiments, may be a bellow or other variable volume container. Once the residue or contaminants within the distillation tank **212** are removed, the valve V.sub.4 is closed. Next valves V.sub.5, V.sub.6 are opened and gaseous CO.sub.2 from the distillation tank **212** flows into the gaseous holding tank **226**. At the end of this part of the process, the distillation tank **212** may be substantially emptied of both liquid/solid wastes which are extracted to the slug tank **222** and gas which is extracted to the gaseous holding tank **226**. In some configurations, the gas may not be extracted until a final purge operation, such as after multiple (e.g., 3 or more) repetitions of dilution. Maintaining the distillation tank pressurized with gas throughout the dilution process may assist in preventing dry ice and/or thermal shock within the distillation tank **212** and/or to ensure and maintain the CO.sub.2 in a liquid state as it is directed into the distillation tank **212**.

[0062] The distillation tank **212** is then re-pressurized. Valves V.sub.3, V.sub.4 may be provided between the distillation tank **212** and the cleaning cavity of the rotary cleaner **208**. It is noted that CO.sub.2 remains a liquid at room temperature between the pressures of about 75 psi to about 1075 psi. Therefore, the pressure difference between rotary cleaner **208** and the distillation tank **212** is such that when the valve V.sub.4 is opened or operated, both the rotary cleaner **208** and the distillation tank **212** must still be both at a pressure where CO.sub.2 can remain a liquid. It will be appreciated that the pressure of the distillation tank **212** will never exceed the pressure of the rotary cleaner **208**, and thus flow of fluids from the distillation tank **212** to the rotary cleaner **208** is prevented. The specific pressures of the system and components thereof may depend on the size and/or volume of the rotary cleaner **208** and distillation tank **212**, the amount of CO.sub.2, the amount of clothing, and/or based on other considerations of the system and/or use thereof.

[0063] To perform the re-pressurization, the valve V.sub.4 is opened or operated and the distillation tank **212** is filled with liquid CO.sub.2 and soil/residue from the rotary cleaner **208**. During this operation, the rotary cleaner **208** and the distillation tank **212** will equalize in pressure (e.g., between 75 psi and 1075 psi) and the CO.sub.2 will remain in liquid form. The valve V.sub.4 is then closed or deactivated. The rotary cleaner **208** will then be brought up to a threshold pressure and/or specified fill level, with clean liquid CO.sub.2, in order to both dilute the contaminated liquid CO.sub.2 and to maintain the pressure range between the rotary cleaner **208** and the distillation tank **212** to ensure that the CO.sub.2 remains in liquid phase through several repetitions of dilution (e.g., multiple repetitions of depressurization/re-pressurization). The depressurization step may then be performed again to remove residue/soil and gas from the distillation tank **212**. Depending on the volume of the distillation tank **212**, the re-pressurization/depressurization step may be repeated multiple times (e.g., three or more), during a cleaning cycle. For example, after repeated cycles, the liquid CO.sub.2 will be diluted enough to proceed with depressurization of the rotary cleaner **208** without risk of re-contamination of the clothes. After the last repetition/cycle, the rotary cleaner **208** may not be repressurized. The distillation tank **212** can then be depressurized/repressurized until all of the liquid CO.sub.2 has been purged and all that is left is the gas. Then, a final purge of the remaining gases may occur, as described herein.

[0064] The slug tank **222**, in combination with the flow control element **224**, can push the residue/soil through the second filter **220** (e.g., a carbon filter), the first filter **216** (e.g., lint filter), and the dispenser **218**. This allows for trace liquid CO_2 that was captured in the soil/residue extraction step. The slug tank **222** may be an optional component that is used to supplement residue/soil removal, and in some embodiments the slug tank may not be employed. Next, valves V_2, V_3 are closed.

[0065] Next, with valves V_{10}, V_6 open, any remaining gaseous CO_2 within the rotary cleaner **208** may be pulled into the gaseous holding tank **226**. This operation may depressurize the rotary cleaner **208**. After depressurization, valves V_{10}, V_6 are closed.

[0066] Next, gaseous CO_2 is reclaimed or recaptured. In this operation, valve V_7 is opened and the compressor **230** is operated. The compressor **230** will pull gaseous CO_2 from the gaseous holding tank **226** and direct the gaseous CO_2 to the condenser **232**. The condenser **232** will convert high pressure gaseous CO_2 from the compressor **230** to the liquid state and ensure all CO_2 being directed back into the holding tank **202** will be in liquid form. It is noted that a pressure balance between the gaseous holding tank **226** and the rotary cleaner **208** is regulated by valve V_7, the compressor **230**, and the holding tank **202**, using pressure gauges, regulators, and the like which may be controlled by the controller **242**. Gas is moved by going from a high pressure area to a low pressure area. To move gas from the rotary cleaner **208** to the gaseous holding tank **226**, pressure must be higher in the rotary cleaner **208** and lower in the gaseous holding tank **226**, and may be performed as described above. To move gas from the gaseous holding tank **226** to the rotary cleaner **208**, the pressure must be higher in the gaseous holding tank **226** than the rotary cleaner **208**, and may be performed as described above. The compressor **230** will be operated to pull gas from the gaseous holding tank **226** to regulate pressure balance, and then the compressor **230** is turned off and valve V_7 is closed.

[0067] At this point the rotary cleaner **208** will stop spinning. The rotary cleaner **208** may then be repressurized to ambient (e.g., pressure of occupied space **240**) to allow for opening of the door **210** and removal of cleaned clothes. Optionally, or if necessary, valve V_{11} may be opened and the fourth pump **236** operated to remove any final amounts of gaseous CO_2 from the rotary cleaner **208** or by fluidly coupling to the vacuum of space, prior to re-pressurization of the rotary cleaner **208** to ambient pressures. The re-pressurization may be achieved by opening valve V_{12} which allows air from the occupied space **240** to enter the empty rotary cleaner **208**. As a final optional step, the compressor **230** may be operated with valves V_7, V_8 open, but all other valves closed, to allow for replenishment of CO_2 into the holding tank **202** from the auxiliary supply **228**.

[0068] Referring now to FIGS. **3**A-**3**B, schematic illustrations of a rotary cleaner **300** in accordance with an embodiment of the present disclosure are shown. FIG. **3**A illustrates a front elevation view of the rotary cleaner **300** and FIG. **3**B illustrates a side elevation view thereof. The rotary cleaner **300** may be incorporated into the various embodiments shown and described herein or with other cleaning systems, as will be appreciated by those of skill in the art. The rotary cleaner **300** is configured to rotate about a central axis **302** and includes an outer housing **304** and an internal drum **306** arranged radially inward from the outer housing **304**. The internal drum **306** may have holes, perforations, openings, a grating, a mesh configuration, or the like, which is selected to allow liquid CO_2 and bonded debris/soil to pass through, but to otherwise hold clothing **308** within the internal drum **306** and to prevent the clothing **308** from contacting the outer housing **304**. The rotary cleaner **300** defines a cleaning cavity **310** defined within the internal drum **306** and annular cavity **312** radially between an external surface of the internal drum **306** and an internal surface of the outer housing **304**. A set of valves V_1-V_7 are provided to enable fluid communication between the rotary cleaner **300** and other components of a cleaning systems, as shown and described herein.

[0069] Arranged within the internal drum **306** and arranged along the central axis **302** is a distillation tank **314**. The distillation tank **314** defines a distillation cavity **316** therein. The distillation tank **314** may be a solid walled structure such that the distillation cavity **316** may be

fluidly separated from the cleaning cavity **310**. However, fluid coupling may be enabled between the distillation cavity **316** and the cleaning cavity **310** by selective operation of valve V.sub.3. The distillation tank **314** may be fixedly mounted to, attached to, or otherwise part of the rotary cleaner **300** such that rotation of the rotary cleaner **300** causes simultaneous rotation of the distillation tank **314**. As noted above, during operation of a rotary cleaner having an internal distillation tank, such as shown in FIGS. **3**A-**3**B, heat may be generated within the cleaning cavity **310** which in turn can heat the distillation tank **314** and any contents therein (e.g., liquid CO.sub.2 may be heated and boiled). In some embodiments, and optional heating element **318** may be provided to apply additional heat to or within the distillation tank **314**. Valves V.sub.1-V.sub.2 may be operated during a cleaning cycle, valves V.sub.3-V.sub.4 may be used during operation of the distillation tank **314** (e.g., pressurization, depressurization, extraction of soil, extraction of liquid or gas CO.sub.2, or the like), and valves V.sub.5-V.sub.7 may be used to depressurize (e.g., by pump or vacuum) or pressurize (e.g., via occupied space air) the rotary cleaner **300**.
[0070] FIG. **4** is a schematic illustration of another configuration of a cleaning system **400** in accordance with an embodiment of the present disclosure. The cleaning system **400**, similar to that of FIGS. **1-2**, is a closed-loop configuration that uses carbon dioxide (CO.sub.2) for cleaning of clothing, although in this configuration there is no distillation tank. The cleaning system **400** includes a holding tank **402** which is configured to supply liquid CO.sub.2 in a flow direction along a flow path **404** from the holding tank **402**. A first pump **406** and a rotary cleaner **408** are provided for receiving both laundry through a door **410** and liquid CO.sub.2 from the holding tank **402**. The rotary cleaner **408** includes an internal drum **412**, as shown and described above. The cleaning system **400** further includes various pumps **414**, **416**, **418**, filters **420**, **422**, an optional dispenser **424** for detergent/deodorizer, a slug tank **426**, a gaseous holding tank **428**, an auxiliary supply **430** of CO.sub.2, a compressor **432**, and a condenser **434**. As shown in this non-limiting configuration, the flow of CO.sub.2 along the flow path **404**, whether liquid or gas, through the cleaning system **400** may be controlled by a set of valves V.sub.1-V.sub.11, which may be controlled by a controller or the like, as described above. Further, the rotary cleaner **408** may be selectively fluidly coupled to an air outlet **436** through the valve V.sub.10 and a pump **418**.
[0071] In operation of the cleaning system **400**, the door **410** may be opened and a user may place soiled clothing within the internal drum **412** of the rotary cleaner **408**. The door **410** may then be closed and seal the cleaning cavity of the rotary cleaner **408**. All valves V.sub.1-V.sub.11 are closed. Valve V.sub.10 or valve V.sub.11 is opened to depressurize the rotary cleaner **408**. For example, valve V.sub.10 may be opened and pump **418** operated to remove air from the rotary cleaner **408**, which can be evacuated to space, extracted and sent to a processing system, or may be otherwise directed through the air outlet **436**. Valve V.sub.10 is then closed.
[0072] With the clothing in the evacuated rotary cleaner **408**, a fill operation is performed. First, as an optional step, valves V.sub.5, V.sub.6 are opened, and gaseous CO.sub.2 from the gaseous holding tank **428** will fill the low pressure rotary cleaner **408**. The pressure will slowly increase within the rotary cleaner **408** to a level of at least 75 psi, to prevent formation of dry ice within the rotary cleaner **408**. Then valves V.sub.5, V.sub.6 are closed. Next, valves V.sub.1, V.sub.2 are opened to permit a fluid connection between the holding tank **402** and the rotary cleaner **408**. A pump **406** is operated to draw liquid CO.sub.2 from the holding tank **402** and direct it into the rotary cleaner **408**. As described above, at first the liquid CO.sub.2 will boil due to the low pressure within the rotary cleaner **408**, but as the rotary cleaner **408** is filled, the pressure will increase to levels sufficient to ensure additional added CO.sub.2 will remain in liquid form (e.g., about 700 psi). Once at pressure, the pump **406** will operate until the rotary cleaner **408** is filled to a specified level, and then the valve V.sub.1 is closed.
[0073] Next, valve V.sub.3 is opened and a pump **414** is operated to cycle the liquid CO.sub.2 from the rotary cleaner **408** through a filter **420** and optional dispenser **424** for adding a detergent, deodorizer, or the like, and then back into the rotary cleaner **408**. The cleaning cycle is then

performed by rotating the rotary cleaner **408** such that liquid CO.sub.2 will be driven radially outward through the clothing and internal drum **412** to collect soil, debris, waste, dirt, and the like from the clothing. When the cleaning cycle is complete, the rotary cleaner **408** will continue to spin, ensuring that the dirty liquid CO.sub.2 does not interact with the clothing (e.g., spins to keep the liquid CO.sub.2 in an annular cavity radially external to the internal drum **412**).

[0074] The dirty CO.sub.2 (e.g., soil and residue that is captured by the liquid CO.sub.2 during the cleaning cycle) may then be removed from the rotary cleaner **408**. Valve V.sub.4, connected to the annular cavity of the rotary cleaner **408**, is opened and a pump **416** is operated to draw residue and soil from the rotary cleaner **408** and into the slug tank **426**. Once the contaminants, soil, residue, etc. is flushed from the rotary cleaner **408**, the valve V.sub.4 is closed. It is noted that at the beginning of the next cycle, the slug tank **426** can be configured to push the residue, soil, debris, etc., through the filters **422**, **420**, and the dispenser **424** to allow for capture any CO.sub.2 that may have been pulled into the slug tank **426**. As an optional step, valves V.sub.1, V.sub.2 may be opened and the rotary cleaner **408** is refilled with liquid CO.sub.2 to a specified fill level and/or pressure threshold. This process may be repeated to dilute the contaminated liquid CO.sub.2. In a variation on the present configuration, similar to FIG. **2**, a single pump (e.g., pump **414**) may be used, and pump **416** may be eliminated. In such a configuration, the valve V.sub.4 may be arranged between the pump **414** and the slug tank **426**. Once the liquid CO.sub.2 has been removed from the rotary cleaner **408**, valve V.sub.4 will close.

[0075] Next, valves V.sub.5, V.sub.6 are opened and used to move any CO.sub.2 within the rotary cleaner **408** to the gaseous holding tank **428** (e.g., by mechanism of pressure differentials or a pump). This operation will depressurize the rotary tank **408**. Once depressurization is complete, valves V.sub.5, V.sub.6 will close.

[0076] The CO.sub.2 gas will then be recaptured by opening valve V.sub.7. The compressor **432** is operated to remove gaseous CO.sub.2 from the gaseous holding tank **428** and sending it through the compressor **432** and condenser **434** and back to the holding tank **402** as liquid CO.sub.2. The compressor **432** is operated until pressure regulation is achieved. The pressure balance between the gaseous holding tank **428** and the rotary cleaner **408** is regulated by valve V.sub.7, the compressor **432**, and the holding tank **402**. Gas is moved by going from a high pressure area to a low pressure area. To move gas from the rotary cleaner **408** to the gaseous holding tank **428**, the pressure must be higher in the rotary cleaner **408** and lower in the gaseous holding tank **428**. To move gas from the gaseous holding tank **428** to the rotary cleaner **408**, the pressure must be higher in the gaseous holding tank **428** than in the rotary cleaner **408**. The compressor **432** will then be deactivated and valve V.sub.7 will close. The rotary cleaner **408** may then stop spinning. The rotary cleaner **408** is repressurized by opening valve V.sub.11 which can introduce ambient air/pressure into the rotary cleaner **408**. Optionally, prior to introduction of air through valve V.sub.11, valve V.sub.10 may be opened to evacuate any excess CO.sub.2 to space (vacuum) and/or a pump **418** may be operated to extract the leftover CO.sub.2. Once the ambient air is introduced into the rotary cleaner **408**, the door **410** may be opened to remove the cleaned clothing. The compressor **432** may also be optionally operated, in combination with opened valves V.sub.7, V.sub.8 to draw additional CO.sub.2 from the auxiliary supply **430** to replenish the CO.sub.2 within the holding tank **402**.

[0077] In the above described configurations, the cleaning systems **100**, **200**, **400** are arranged as closed-loop systems. However, such closed-loop arrangement is not required and various embodiments of the present disclosure may be open-loop. For example, with reference to FIG. **5**, a schematic diagram of an open-loop cleaning system **500** in accordance with an embodiment of the present disclosure is shown. The cleaning system **500** is arranged substantially similar to that shown and described with respect to FIG. **2**, but in an open-loop arrangement.

[0078] The cleaning system **500** includes a holding tank **502** which is configured to supply liquid CO.sub.2 in a flow direction along a flow path **504** from the holding tank **502**. A first pump **506** and a rotary cleaner **508** are provided for receiving both laundry through a door **510** and liquid

CO.sub.2 from the holding tank **502**. The rotary cleaner **508** includes an internal drum, as shown and described above, and an internal distillation tank **512**. The cleaning system **500** further includes various pumps **514**, **516**, **518**, filters **520**, **522**, an optional dispenser **524** for detergent/deodorizer, a slug tank **526**, a gaseous holding tank **528**, an auxiliary supply **530** of CO.sub.2, a compressor **532**, and a condenser **534**. As shown in this non-limiting configuration, the flow of CO.sub.2 along the flow path **504**, whether liquid or gas, through the cleaning system **500** may be controlled by a set of valves V.sub.1-V.sub.13, which may be controlled by a controller or the like, as described above. Further, the rotary cleaner **508** may be selectively fluidly coupled to an air outlet **536** through valve V.sub.11 and a pump **518**. In this non-closed-loop configuration, CO.sub.2 that is pulled from the rotary cleaner **508** may be collected in a separate system and/or ejected to vacuum or otherwise disposed of through valve V.sub.7, as shown.

[0079] In operation of the cleaning system **500**, a user may open the door **510** and place clothes within the internal drum of the rotary cleaner **508**. The door **510** is then closed and sealed. All valves V.sub.1-V.sub.13 are closed, and then valve V.sub.11 is opened to evacuate the rotary cleaner **508** of air, and then valve V.sub.11 is closed. The rotary cleaner **508** is then filled with liquid CO.sub.2 in a similar fashion as described above by opening valves V.sub.1, V.sub.2 and pumping CO.sub.2 from the holding tank **502** into the rotary cleaner **508** until a threshold pressure is reached and liquid CO.sub.2 will fill the rotary cleaner **508** to a specified level and/or pressure level. A cleaning cycle with valves V.sub.2, V.sub.3 open and valve V.sub.1 closed may be performed with pump **514** pumping the liquid CO.sub.2 through filter **520** and dispenser **524**, which can introduce detergent and/or deodorizers and the like. Once the cleaning is complete, the rotary cleaner **508** will be rotationally driven and continue spinning, as described above. Next, the distillation tank **512**, which was warmed during the cleaning operation, can be depressurized by opening valve V.sub.4 and operating pump **516** or performed as otherwise described herein (e.g., with respect to FIG. **2**). In a variation on the present configuration, similar to FIG. **2**, a single pump (e.g., pump **514**) may be used, and pump **516** may be eliminated. In such a configuration, an additional valve may replace the pump **516** between the valve V.sub.4, the pump **514**, and the slug tank **526**. Further, in some such configurations, an additional fluid connection may be provided between valves V.sub.3, V.sub.4. The contents of the distillation tank **512** may be directed into the slug tank **526**. Once the contaminants withing the distillation tank **512** are removed, the pump **516** may be turned off and the valve V.sub.4 is closed. Valve V.sub.5 may then be opened and closed to vent gaseous CO.sub.2 and then reseal the distillation tank **512**.

[0080] The distillation tank **512** will then be repressurized by opening valve V.sub.13, which is connected to the annular cavity of the rotary cleaner **508**, to fluidly couple and equalize the pressure between the rotary cleaner **508** and the distillation tank **512**. When valve V.sub.13 is opened, liquid CO.sub.2 will enter the distillation tank **512** from the rotary cleaner **508**. As the liquid CO.sub.2 flows into the distillation tank **512**, it will carry any soil, debris, etc. that was picked up from the clothing during the cleaning cycle. In a non-limiting example, the distillation tank **512** and the rotary cleaner **508** may equalize around between 75 psi and 1075 psi, which ensures that the CO.sub.2 remains in liquid form. Valve V.sub.13 is then closed, sealing the distillation tank **512** from the rotary cleaner **508**. In some configurations, depressurization and re-pressurization of the distillation tank may be performed multiple times to remove residue/gas. The slug tank **526** can then push the residue through the filters **522**, **520**, and the dispenser **524**, and trace liquid CO.sub.2 may be recovered during the next cleaning cycle. Valves V.sub.2, V.sub.3 are then closed. Valve V.sub.6 can then be opened to remove any CO.sub.2 gas still presented within the rotary cleaner **508**, and then closed again. At this point the cleaning cycle and processing is complete, and the rotary cleaner **508** may be pressurized, as described above, and the clean clothes may be removed.

[0081] Referring now to FIG. **6**, another open-loop configuration of a cleaning system **600** in accordance with an embodiment of the present disclosure is shown. The cleaning system **600** includes a holding tank **602** which is configured to supply liquid CO.sub.2 in a flow direction along a flow path **604** from the holding tank **602**. A first pump **606** and a rotary cleaner **608** are provided for receiving both laundry through a door **610** and liquid CO.sub.2 from the holding tank **602**. The rotary cleaner **608** includes an internal drum **612**, and the cleaning system **600** does not include a distillation tank. The cleaning system **600** further includes various pumps **614**, **616**, **618**, filters **620**, **622**, an optional dispenser **624** for detergent/deodorizer, a slug tank **626**, a gaseous holding tank **628**, an auxiliary supply **630** of CO.sub.2, a compressor **632**, and a condenser **634**. As shown in this non-limiting configuration, the flow of CO.sub.2 along the flow path **604**, whether liquid or gas, through the cleaning system **600** may be controlled by a set of valves V.sub.1-V.sub.11, which may be controlled by a controller or the like, as described above. Further, the rotary cleaner **608** may be selectively fluidly coupled to an air outlet **636** through valve V.sub.10 and a pump **618**. In this non-closed-loop configuration, CO.sub.2 that is pulled from the rotary cleaner **608** may be collected in a separate system and/or ejected to vacuum or otherwise disposed of through valve V.sub.6, as shown.

[0082] In operation of the cleaning system **600**, a user may open the door **610** and place clothes within the internal drum **612** of the rotary cleaner **608**. The door **610** is then closed and sealed. All valves V.sub.1-V.sub.11 are closed, and then valve V.sub.10 is opened to evacuate the rotary cleaner **608** of air, and then valve V.sub.10 is closed. The rotary cleaner **608** is then filled with liquid CO.sub.2 in a similar fashion as described above by opening valves V.sub.1, V.sub.2 and pumping CO.sub.2 from the holding tank **602** into the rotary cleaner **608** until a threshold pressure is reached and liquid CO.sub.2 will fill the rotary cleaner **608** to a specified level and/or pressure level. A cleaning cycle with valves V.sub.2, V.sub.3 open and valve V.sub.1 closed may be performed with pump **614** pumping the liquid CO.sub.2 through filter **620** and dispenser **624**, which can introduce detergent and/or deodorizers and the like. Once the cleaning is complete, the rotary cleaner **608** will be rotationally driven and continue spinning, as described above. Next, residue is removed by opening valve V.sub.4, which is connected to the annular cavity of the rotary cleaner **608**, and pump **616** is operated to direct the residue from the rotary cleaner **608** into the slug tank **626** or performed as otherwise described herein (e.g., with respect to FIG. **2**). In a variation on the present configuration, similar to FIG. **2**, a single pump (e.g., pump **614**) may be used, and pump **616** may be eliminated. In such a configuration, the valve V.sub.4 may be arranged between the pump **614** and the slug tank **626**. Once contaminants are flushed out of the rotary cleaner **608**, the valve V.sub.4 is closed. At the beginning of the next cleaning cycle any residue within the slug tank **626** can be pushed through the filters **622**, **620** and conditioned at the dispenser **624**, to allow for recovery of any remaining liquid CO.sub.2 that was held in the slug tank **626**. Valves V.sub.5, V.sub.6 can be opened to vent any remaining gaseous CO.sub.2 in the rotary cleaner **608** into space or otherwise may be pumped out of the cleaning system **600**. The rotary cleaner **608** is then repressurized by opening valve V.sub.11, and the clean clothes may be removed.

[0083] As noted, each of the cleaning systems **500**, **600** is an open-loop configuration. That is, the gaseous CO.sub.2 that is extracted from the rotary cleaner **508**, **608** after the cleaning cycle may be drawn out and disposed by opening a respective valve and venting to atmosphere, vacuum, or the like, or the CO.sub.2 may otherwise be extracted for other uses or systems. In these configurations, the CO.sub.2 is not actively reclaimed within the cleaning systems **500**, **600**. However, additional CO.sub.2 may be introduced to replenish the respective cleaning systems **500**, **600** from auxiliary supplies **530**, **630**, with the compressors **532**, **632** and the condensers **534**, **634** used to convert gaseous CO.sub.2 to a liquid state prior to being directed into the respective holding tanks **502**, **602**. The cleaning systems **500**, **600** may be configured for use in environments where CO.sub.2 may be extracted from a local atmosphere or other local source (e.g., on a planetary body or the like, such as the atmosphere of Mars with 95% CO.sub.2). In such situations, a closed-loop system may not be necessary. As such, the complexity of the cleaning system may be reduced by eliminating the closed-loop nature of the embodiments of FIGS. **1**, **2**, and **4**.

[0084] In view of the above, embodiments of the present disclosure are directed to cleaning systems for clothing and other fabrics (e.g., bedding, towels, etc.) that can be used in microgravity environments. The cleaning systems employ a rotary cleaner and liquid CO_2 as a cleaning agent that will bind to soil, debris, waste, dirt and the like and pull it off the clothing as the rotary cleaner rotates. The use of an internal drum that is porous to liquid CO_2 (with bound contaminants) but solid with respect to the cloth to be cleaned provides for a low demand from conventional resources. That is, the cleaning systems described herein eliminate the need for water to be used in a cleaning process for clothing. This provides benefits as compared to conventional systems which require water which is a scarce or non-existent resource in certain situations, such as during space travel and exploration. Since water is such a valuable resource, eliminating the need for water in cleaning and sanitation operations can provide immense benefits.

[0085] Although only a limited number of configurations have been shown and described, both closed-loop and open-loop, it will be appreciated that variations thereon are within the scope of the present disclosure. For example, various additional components or elements may be incorporated into the cleaning systems described herein. In one such example, additional deodorizers or the like may be employed because CO_2 may not sufficiently remove scents and odors and/or surfactants that may be introduced to remove stains or the like. Additionally, the cleaning systems described herein may be integrated into larger systems, or coupled thereto. For example, the condensers used to convert gaseous CO_2 to liquid CO_2 (e.g., downstream from the compressors) may be coupled to various other systems that operate to provide a material or fluid for heat pick up and cause the phase change of the CO_2.

[0086] Furthermore, although shown and described with specific valve configurations and operations, it will be appreciated that other valve configurations may be implemented in one or more of the above described embodiments, without departing from the scope of the present disclosure. For example, various embodiments may have fewer or greater valves than those shown, and the illustrated valve configurations are merely for explanatory and illustrative purposes. Additionally, although described as valves, these components may take the form of other structures, which are used to control fluid flow, as will be appreciated by those of skill in the art. Furthermore, in some embodiments, two or more of the illustrated and/or described valves may be combined into a single valve or valve assembly.

[0087] Additionally, it will be appreciated that sensors may be distributed throughout the various configurations and operably connected to a controller. The sensors may include temperature sensors, fluid sensors, flow sensors, pressure sensors, valve-state sensors, CO_2 liquid quality sensors (e.g., to detect debris/soil, etc.), and the like. Furthermore, various control elements may be employed, such as actuators, electronic controllers, motors, generators, electrical power supply, and the like may be provided from the station or craft on which the cleaning systems are employed.

[0088] Advantageously, embodiments of the present disclosure provide for improved cleaning systems for laundry and the like that may be used in microgravity environments. Benefits of the embodiments described herein include, for example and without limitation, elimination of the use of water for laundry and cleaning operations onboard space stations or space craft where water is a limited and valuable resource. Accordingly, water consumption for sanitation may be reduced by use of cleaning systems as described herein. Furthermore, advantageously, because of the manipulation of pressure to ensure the phase state of CO_2, the systems described herein may be wholly or primarily operated at room temperature, and no cooling is required to maintain the CO_2 in liquid state. In some configurations, additional heat may be input into the system to ensure boiling of liquid CO_2 to convert the liquid to a gas state for the purpose of cleaning and removing of soil and the like that is captured by the liquid CO_2.

[0089] Additionally, due to the CO_2 being removed from the clothing and evaporated, the clothes that are cleaned within the cleaning systems described herein may be dry when they are

removed from the rotary cleaner. As such, no additional heating is required to perform a drying operation/function. Furthermore, the use of CO.sub.2 may be gentler on clothing as compared to washing with water, and thus the useful life of the clothing may be extended beyond that possible with water-based cleaning solutions.
[0090] The use of the terms “a”, “an”, “the”, and similar references in the context of description (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or specifically contradicted by context. The modifier “about” used in connection with a quantity is inclusive of the stated value and has the meaning dictated by the context (e.g., it includes the degree of error associated with measurement of the particular quantity). All ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. As used herein, the terms “about” and “substantially” are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, the terms may include a range of ±8% of a given value or other percentage change as will be appreciated by those of skill in the art for the particular measurement and/or dimensions referred to herein.
[0091] While the present disclosure has been described in detail in connection with only a limited number of embodiments, it should be readily understood that the present disclosure is not limited to such disclosed embodiments. Rather, the present disclosure can be modified to incorporate any number of variations, alterations, substitutions, combinations, sub-combinations, or equivalent arrangements not heretofore described, but which are commensurate with the scope of the present disclosure. Additionally, while various embodiments of the present disclosure have been described, it is to be understood that aspects of the present disclosure may include only some of the described embodiments. Accordingly, the present disclosure is not to be seen as limited by the foregoing description but is only limited by the scope of the appended claims.

Claims

1. A carbon dioxide-based cleaning system comprising: a holding tank containing a supply of liquid carbon dioxide (CO.sub.2); a rotary cleaner having an internal drum arranged therein, the rotary cleaner fluidly coupled to the holding tank to receive liquid CO.sub.2 therefrom, the rotary cleaner configured to output CO.sub.2 with contaminants after a cleaning cycle; a compressor arranged to receive gaseous CO.sub.2 and increase a pressure of the gaseous CO.sub.2; and a condenser arranged to receive the increased pressure gaseous CO.sub.2 from the compressor and convert the gaseous CO.sub.2 into liquid CO.sub.2 and direct the liquid CO.sub.2 into the holding tank.
2. The cleaning system of claim 1, wherein the output CO.sub.2 with contaminants is directed from the rotary cleaner to the compressor after the cleaning cycle.
3. The cleaning system of claim 2, further comprising a distillation tank arranged between the rotary cleaner and the compressor, the distillation tank configured to receive at least one of gaseous and liquid CO.sub.2 from the rotary cleaner and separate contaminants from the CO.sub.2 and supply clean gaseous CO.sub.2 to the compressor.
4. The cleaning system of claim 3, wherein the distillation tank is coaxially arranged within the rotary cleaner.
5. The cleaning system of claim 4, wherein the distillation tank comprises a valve configured to selective fluidly couple a distillation cavity defined within the distillation tank with a cleaning cavity defined within the rotary cleaner and external to the distillation tank.
6. The cleaning system of claim 3, further comprising a heating element arranged to provide heat to the distillation tank.
7. The cleaning system of claim 3, further comprising a slug tank arranged to receive CO.sub.2 and contaminants captured by the CO.sub.2 from the distillation tank.

8. The cleaning system of claim 1, wherein the internal drum is porous to liquid CO.sub.2 and contaminants carried by the liquid CO.sub.2 but prevents clothing from passing out of the internal drum.
9. The cleaning system of claim 1, wherein the rotary cleaner comprises a cleaning cavity defined within the internal drum and an annular cavity is defined radially between an external surface of the internal drum and an internal surface of an outer housing of the rotary cleaner.
10. The cleaning system of claim 1, further comprising a valve arranged on the rotary cleaner and configured to selectively open to evacuate gas from a cleaning cavity of the rotary cleaner.
11. The cleaning system of claim 10, further comprising a pump associated with the valve, the pump configured to provide a motive force to draw the gas from the cleaning cavity of the rotary cleaner.
12. The cleaning system of claim 1, further comprising a filter and pump connected to the rotary cleaner, wherein the pump is configured to pull liquid CO.sub.2 from the rotary cleaner and pass it through the filter and then redirect the liquid CO.sub.2 back into the rotary cleaner.
13. The cleaning system of claim 12, further comprising a dispenser arranged along a flow path through the pump and the filter, wherein the dispenser is configured to apply at least one of a deodorizer and a detergent to the liquid CO.sub.2 prior to being redirected back into the rotary cleaner.
14. The cleaning system of claim 1, further comprising an auxiliary supply of CO.sub.2 arranged upstream from the compressor, the auxiliary supply of CO.sub.2 configured to provide additional CO.sub.2 into the system to replenish losses from the holding tank.
15. The cleaning system of claim 1, further comprising a slug tank arranged to receive CO.sub.2 and contaminants captured by the CO.sub.2 from the rotary cleaner.
16. The cleaning system of claim 1, further comprising a valve configured to fluidly couple a cleaning cavity of the rotary cleaner with an ambient environment.
17. The cleaning system of claim 16, wherein the ambient environment is defined within a spacecraft.
18. The cleaning system of claim 16, wherein the ambient environment is defined within a space station.
19. The cleaning system of claim 16, wherein the ambient environment is defined within a station on a non-Earth celestial object.
20. The cleaning system of claim 1, wherein the holding tank is a bellows tank arranged to maintain the liquid CO.sub.2 under pressure and in a liquid state.
